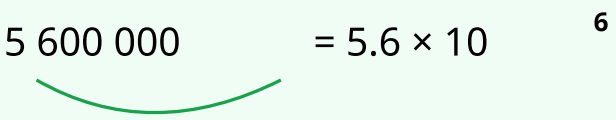
Standard Form — Writing and Converting

AQA · KS4 Year 9–10

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE



Standard form: A × 10ⁿ with 1 A < 10.

COMMON MISTAKES

A must satisfy 1 A < 10. So 36 × 10^4 is wrong; correct: 3.6 × 10^5.

Positive power = big; negative power = small.

Count places carefully — 0.00047 needs 4 moves 4.7 × 10^{-4}.

0.003 = 3 × 10^{-3}, NOT 3 × 10^3.

METHOD

1. Move decimal so 1 A < 10
2. Count places moved = power n
3. Big number (1) positive n
4. Small number (< 1) negative n
5. Check by converting back

WORKED EXAMPLE

Write 0.000 47 in standard form.
Step 1 — Move decimal 4 places: 4.7
Step 2 — Original < 1, so n = -4
Answer: 4.7 × 10^{-4}

VISUAL AID

Place-value slide showing the decimal point moving left for big numbers (positive power) and right for small numbers (negative power). Arrow diagram: 5,600,000 5.6 (6 places left) = 5.6 × 10^6. And 0.0034 3.4 (3 places right) = 3.4 × 10^{-3}.

1. (2 marks)
Standard form: a) 5,000 b) 340,000 c) 72,000,000 d) 810

2. (2 marks)
Standard form: a) 0.003 b) 0.00056 c) 0.0000012 d) 0.47

3. (2 marks)
To ordinary: a) 3.2 × 10^4 b) 7.05 × 10^6 c) 1.8 × 10^{-3} d) 9.1 × 10^{-5}

4. (3 marks)
Not standard form? Fix: a) 5.6 × 10^3 b) 36 × 10^4 c) 0.8 × 10^{-2} d) 7 × 10^9

5. (3 marks)
Hydrogen atom mass 0.00000000000000000000000167 kg. Standard form?

6. (3 marks)
Earth–Sun = 1.5 × 10^8 km; Earth–Mars = 2.3 × 10^8 km. a) Sun distance ordinary. b) Closer? c) Difference in standard form.

7. (4 marks)
Order smallest first: 3.2 × 10^{-3}, 8.1 × 10^{-4}, 5.5 × 10^{-3}, 2.9 × 10^{-2}. Strategy?

8. (5 marks)
Bacterium = 2 × 10^{-6} m; virus = 3 × 10^{-8} m. a) How many times longer? b) 500 bacteria end-to-end? c) Viruses in 1 mm?